

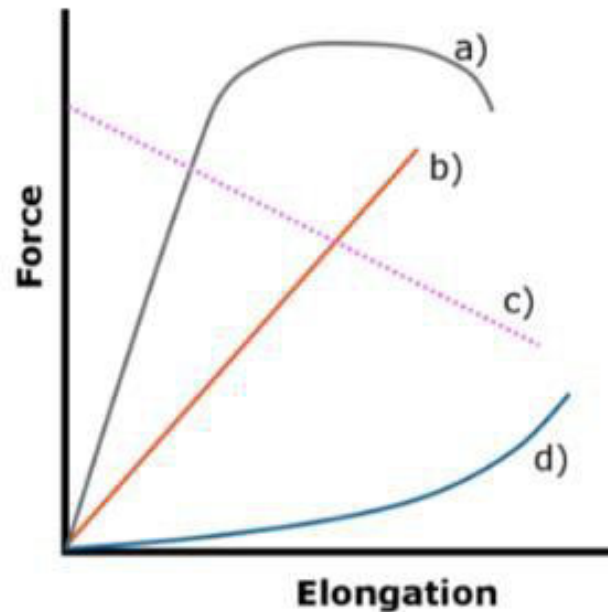
APS110 Tutorial #1

Daniel Brlas

Email: daniel.brlas@mail.utoronto.ca

Practice Question 1

1. If you were to hang four weights of different mass on a spring and plot force applied versus the elongation of the spring, which of the following curves would you expect to see?



A and B are both correct, A shows the elastic and plastic behaviour of the material and B is only the elastic region

Practice Question 2

2. You are given a sample of a metal bar with dimensions of $w: 5 \text{ mm} \times h: 4 \text{ mm} \times l: 20 \text{ cm}$, and you need to test whether or not the material can withstand a stress of 80 MPa . Assuming you apply a load along the long axis, what load would you need to apply to achieve this stress?

a) 1600 N

b) $1\,600\,000 \text{ N}$

c) $4\,000\,000 \text{ N}$

d) 4000 N

$$\sigma = \frac{F}{A}$$

$$F = \sigma A$$

$$F = (80 \times 10^6) \left((5 \times 10^{-3}) \times (4 \times 10^{-3}) \right)$$

$$F = 1600 \text{ N}$$

Practice Question 3

Use the load-elongation characteristics of a cylindrical rod in tension shown in the following table to compute the engineering stress vs. engineering strain and draw what the curve should look like.

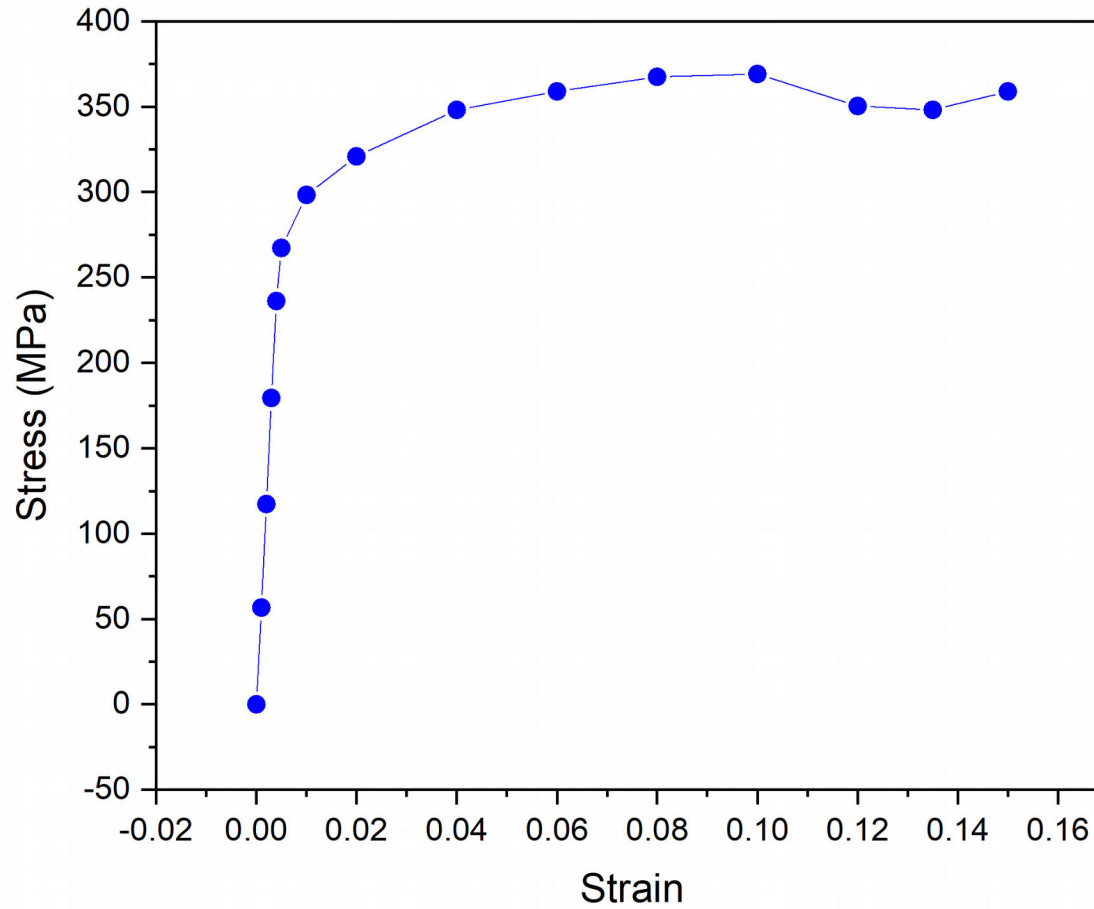
Load (N)	Length (mm)
0	50.8
7300	50.851
15100	50.902
23100	50.952
30400	51.003
34400	51.054
38400	51.308
41300	51.816
44800	52.832
46200	53.848
47300	54.864
47500	55.88
46100	56.896
44800	57.658
46200	58.42

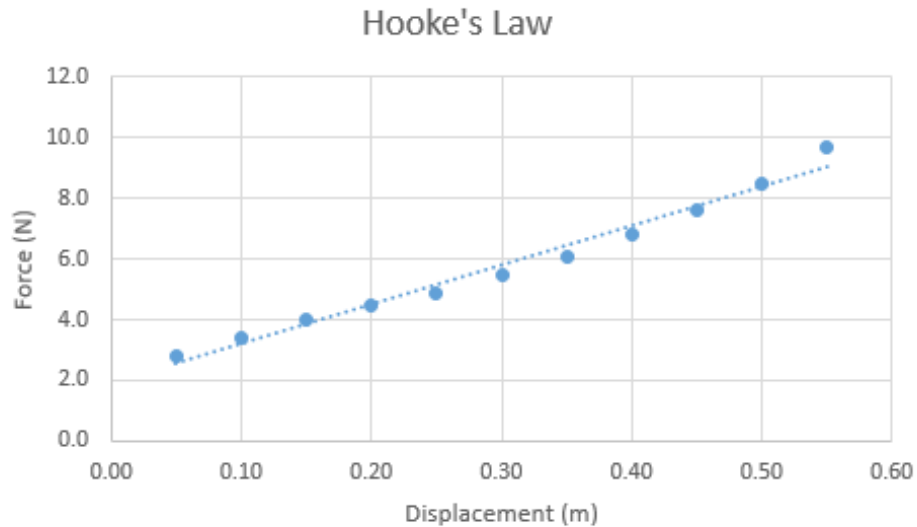
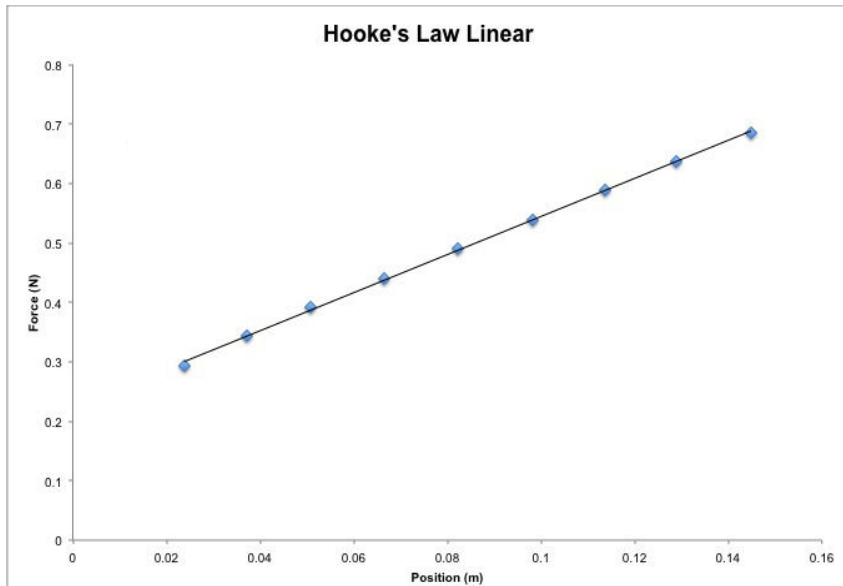
Rod diameter = 12.8mm

$$A = \pi \left(\frac{1.29e-4}{2} \right)^2 = 1.29e-4$$

Use $\sigma = \frac{F}{A}$ and $\varepsilon = \frac{\Delta l}{l_0}$

Stress (Pa)	Strain
0	0
56758808	0.001004
1.17E+08	0.002008
1.8E+08	0.002992
2.36E+08	0.003996
2.67E+08	0.005
2.99E+08	0.01
3.21E+08	0.02
3.48E+08	0.04
3.59E+08	0.06
3.68E+08	0.08
3.69E+08	0.1
3.58E+08	0.12
3.48E+08	0.135
3.59E+08	0.15





Practice Quiz

1. A spring is stretched due to a 50 N load. Calculate the elongated length in cm if the spring constant is 1250 N/m.
2. A large rubber band ball is created with fresh rubber elastics and the last bands are stretched with the equivalent of a 5 cm displacement from the resting position. Over a few months their spring constant decreases by 25% (due to dryness, overstretch, etc.). If the initial spring constant was 17.4 N/m, calculate the load in the rubber band after that period of time.
3. The roof of a gazebo weighs 600 kg and is held by 6 wood posts. The posts are 2 m in height and have a 10x10 cm cross-section. What is the stress on each post supporting the roof?
4. Calculate the approximate spring constant for the two plots on the left. Which of these springs is stiffer?

1. Hooke's law: $F = -kx$
 $x = [50\text{N} / (1250\text{ N/m})] = 0.04\text{m} = 4\text{cm}$
2. Initial load: $F = -kx = [17.4\text{ N/m} * 0.05\text{m}] = 0.87\text{ N}$
Decrease of 25% of k means decrease of 25% of load: 0.65 N
3. Weight on each post: $600\text{kg} / 6 = 100\text{kg}$
Stress on each post: $\sigma = F/A = (100\text{kg} * 9.8\text{m/s}^2) / (0.1\text{m} * 0.1\text{m}) = 98\,000\text{ N/m}^2 = 98\text{ kPa}$
4. Spring constant is the slope of the line, 3.2 N/m for one and 13 N/m for the other.
The stiffer spring has the higher spring constant (more force required to move same distance)

